

# THEMIS: Supervisor Recommendation from a University's Publications and Thesis Postings

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## Abstract

A student looking for a thesis supervisor at a large university has to assemble the answer from a publication repository, researcher profiles and department pages, which are organised differently and rarely say who supervises what. We present THEMIS, a system that recommends supervisors at the University of Zurich from 214,756 publication records in the university's repository and 695 thesis postings scraped from 103 department pages. THEMIS embeds both sources with BGE-M3, queries them separately, credits each retrieved document to the researchers it names, and returns a ranked list of people with links to the documents that support each recommendation. The university's chat assistant can call it as a Model Context Protocol tool. Working with the real corpus exposed two problems: the sources name the same people differently, and they produce similarity scores on different scales, which leaves almost no room for a single abstention threshold. On 900 queries generated from the researchers' own documents, a gold supervisor appears in the top five for 83.2% of queries (MRR 0.730). Two exploratory probes point to weaknesses. Rankings often did not favour a researcher's own topic over broader or neighbouring ones, and queries generated with the student's background scored lower than topic-only queries.

## CCS Concepts

• **Information systems** → **Expert search**; *Recommender systems*; *Evaluation of retrieval results*.

## Keywords

expert finding, supervisor recommendation, dense retrieval, entity resolution, Model Context Protocol

[author list and order to be agreed by the team]

## 1 Introduction

Suppose a student at the University of Zurich (UZH) wants to write a master's thesis on detecting misinformation with language models. The institutional repository, ZORA,<sup>1</sup> records who has published on the topic but not who supervises theses. Open thesis topics are advertised on department and research-group pages, each in its own format, and many name no supervisor. Unless the student already knows someone in the right group, finding a supervisor means reading through several of these sources by hand.

THEMIS (THEsis Matching and Information System) takes a free-text description of a student's interests and returns the researchers whose work matches it, each with the publications and thesis postings that support the match. The two sources play different roles.

Publications show sustained expertise, and postings show topics that a group has advertised. THEMIS keeps the two apart during retrieval and reports which source each recommendation rests on.

The system is built to run inside askUZH, the university's chat assistant, over the Model Context Protocol (MCP) [12]; that integration is not yet deployed. THEMIS returns structured evidence and leaves the answer to the caller, and it also runs from the command line. Its intended users are students choosing a supervisor, and the demonstration should also interest researchers working on expert finding over institutional data. We contribute:

- a demonstrated workflow in which a student's request is matched against publications and thesis postings separately, and each recommended person comes with links to the evidence behind it, exposed to assistants through MCP (Sections 3 and 4);
- two design decisions forced by the real corpus, strict resolution of people across the sources and abstention thresholds set per source, each with its measured limits (Sections 3.3 and 3.4);
- an evaluation on 900 generated queries, with two exploratory probes that expose weaknesses (Section 5).

[links to the demo video and the code repository; both venues require a demo link]

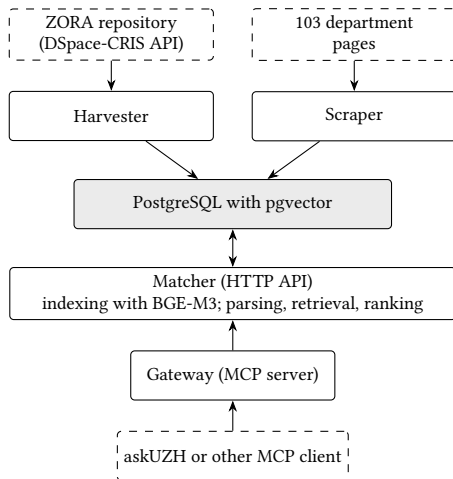
## 2 Related Work

Recommending supervisors is a form of expert finding, the task of ranking people by their knowledge of a topic. The TREC Enterprise track established it as a shared benchmark [5], and Balog et al. formalised two approaches [1, 2]. One builds a profile for each candidate and ranks the profiles; the other ranks documents and aggregates their scores to the people associated with them. THEMIS takes the document-centric approach with the simplest aggregation, the maximum score over a person's retrieved documents.

The closest applied setting is reviewer assignment, which matches submitted papers to reviewers through the reviewers' own publications [3, 11]. In supervisor recommendation the user is a student with a loosely formed interest rather than an author with a finished paper. Thesis postings add a second kind of evidence, topics a group has advertised, which publications cannot provide. Not every matching person is eligible either, since a co-author from another institution may know the topic but cannot supervise a UZH thesis.

THEMIS can also write a short answer from the retrieved evidence, in the manner of retrieval-augmented generation [9]. In the assistant integration, however, its main output is structured evidence, and the caller writes the answer.

<sup>1</sup><https://www.zora.uzh.ch>



**Figure 1: Components of THEMIS.** Dashed boxes are external. The harvester and the scraper write source records, the matcher embeds them and answers queries, and the gateway exposes the matcher to assistants as MCP tools.

### 3 System

Figure 1 shows the components. Each runs as a separate process in its own container image.

#### 3.1 Data

Publications come from ZORA through its DSpace-CRIS REST API. The harvester stores 214,756 records. The repository’s authority links separate authors registered at UZH from external co-authors; 53,545 publications have at least one registered UZH author, and these carry 2,942 distinct UZH author names. Incremental runs fetch newly accessioned records.

Thesis postings come from a scraper over 103 curated department pages. Each source has a declarative specification that locates listings and fields on the page, and extraction from these specifications is deterministic. A language model drafts a specification when a source is added, which a person then approves, and summarises application-procedure pages and, for some sources, linked topic PDFs. It is also the fallback when a specification matches nothing, and records produced that way are flagged for review. Fetched pages are cached, so extraction can be rerun without contacting the sites, and every request carries a contact address. The index holds 695 postings, which record, where the page gives them, the named supervisors, the degree levels a topic is open to, and its status (open, pending, assigned or private).

#### 3.2 Indexing and Retrieval

BGE-M3 [4] embeds each publication from its title, its abstract where present, and its keywords, and each posting from its text (1024 dimensions, input truncated at 1024 tokens). The vectors are stored in PostgreSQL with pgvector [8] and searched by cosine similarity over an HNSW index [10]. A content hash per document means that re-indexing embeds only new or changed records.

A query is first parsed into topics, keywords, an optional degree level and an optional department, by a rule-based parser or, when one is configured, by a language model. The topics and keywords are embedded as one query vector, and THEMIS runs two retrievals, one over publications and one over postings. Degree level exists only on postings, so a single query filtered on “master’s thesis” would drop every publication. Postings marked as assigned or private are excluded; pending postings and postings without a status remain retrievable.

Each retrieved document is credited to people. A posting credits its named supervisors, and a publication credits its registered UZH authors. Only a publication with no registered UZH author credits all of its authors, so an external co-author of a UZH paper never receives credit. A person’s score is the highest similarity among their credited documents, and the source of that document is kept with the score. By default, researchers registered at UZH rank above external researchers. Within each group candidates are ordered by raw similarity, even though the two sources score on different scales (Section 3.4).

#### 3.3 Resolving People Across Sources

The two sources write names differently. ZORA stores “Müller, Anna”, while a department page may write “Anna Müller” or “Prof. Dr. Anna Müller”. Grouped on the raw string, none of the 403 distinct supervisor names on postings matched any of the 2,942 UZH author names in ZORA.

THEMIS maps each name to a key formed from the first given name and the family name, after Unicode normalisation and with academic titles removed. Publications supply the reference keys, because ZORA’s comma marks where a name divides, and names from postings are resolved against them. The rule favours precision over recall. The full first given name must agree, and a family name alone never suffices, because a false merge would present another person’s publications to a student as evidence for a supervisor.

With this rule, 103 of the 403 supervisor names (25.6%) resolve to a ZORA author. We detected no false merge, although no rule on names can separate two people who share both a first and a family name. Most of the 300 unresolved names do not appear among ZORA’s registered UZH authors at all, because thesis supervisors are often doctoral students, postdocs or staff of partner institutes. At the default depth of five results per source the resolution rarely changes a result. Over five probe queries, none of the 25 returned matches combined the two sources.

#### 3.4 Abstention

When nothing in the corpus fits a request, the system should say so rather than present the nearest researchers as a recommendation. When a language model writes the answer, THEMIS first drops every candidate whose score falls below the threshold for that score’s source. If none remains, it returns a fixed reply that there is no strong match and names the closest candidate as a long shot, without calling the model. The structured tool and the offline template answer do not apply this filter. They return the scores and their sources, and the MCP tool does not yet tell the calling assistant which thresholds to use.

**Table 1: Best retrieved match (cosine similarity) for the probe queries on the full index, and the provisional threshold per source. Controls are presumed out of domain.**

| Source       | Controls (highest) | On-topic (lowest) | Threshold |
|--------------|--------------------|-------------------|-----------|
| Publications | 0.542              | 0.605             | 0.57      |
| Postings     | 0.431              | 0.564             | 0.48      |

We set the thresholds from nine probe queries on the full index. Five were on-topic queries spread across faculties, and four were controls that we presume to be out of domain, everyday questions such as how to proof sourdough. In corpus-wide scans for the five on-topic queries, no document scored below 0.115, which is consistent with the anisotropy of contextual embeddings [6]. In both sources, every control’s best retrieved match scored below every on-topic query’s best retrieved match (Table 1). The sources separate at different levels, however. A single threshold would have to fall in an interval only 0.022 wide, and any value in it discards some of the top five postings for on-topic queries. THEMIS therefore uses provisional thresholds per source, 0.57 for publications and 0.48 for postings. One gap remains. A person found in both sources is judged by whichever of their hits scored higher, so a publication just below its threshold can hide a posting that would pass.

### 3.5 Serving and Integration

Serving is split into two processes. The matcher holds the embedding model, the database connection and the language-model client, and exposes an HTTP API for matching, recommendation and index status. The gateway is a thin MCP server over streamable HTTP with two tools. `find_researchers` returns the ranked researchers and their evidence as structured data, and `recommend_supervisors` returns a short written answer. Keeping the model out of the gateway lets it fit the memory quota of the university’s Kubernetes namespace.

When THEMIS writes the answer itself, the language model sees only the retrieved candidates and their listed work. It is instructed to name no one else and to say plainly when a candidate fits only in part. An early run showed that the prompt must also omit information the system does not have. Given a line stating that a researcher had no open position in our data, the model wrote that the researcher was not accepting students, which the data cannot support. The prompt now leaves out the posting line when there is none.

## 4 Demonstration

Visitors use THEMIS through a generic MCP client connected to the gateway, which shows each tool call and its structured result, or through the command line. A visitor describes a thesis interest in their own words, for example a master’s thesis on misinformation detection with language models, and receives researchers in rank order. Each comes with a department, a score, the source of that score, and the publications and postings behind the match, linked to ZORA and to the department page. Adding “master’s thesis” restricts postings to topics open at that level and leaves publications in place.

Visitors can then probe the design decisions. The source field shows whether a recommendation rests on publications or on a posting. With a language model configured, a request that no research group covers, such as the sourdough control of Section 3.4, falls below both thresholds and receives the fixed no-match reply. Setting the written answer from `recommend_supervisors` beside the structured list lets visitors check that every person it names appears in the list. **[record one full-index session with its latency, choose the demonstration queries from it, and decide how to show real researchers’ names]**

## 5 Evaluation

### 5.1 Benchmark

No collection of real student queries with judged supervisors exists for UZH, so we generated one. We sampled 150 researchers with a verified repository identity and at least three publications with abstracts, and 150 postings. KeyBERT [7], with a multilingual sentence encoder [14], extracted keywords from up to 20 of each researcher’s publications with abstracts and from the text of each posting. A language model then phrased up to 2, 4 or 8 of these keywords as a student’s request, which gives three levels of detail per target and 900 queries. **[name the model that phrased the queries; the benchmark manifest does not record it]** The researchers behind the source documents are the gold answer. Each publication query has one, and 60 of the posting queries have several, because their postings name more than one supervisor.

We query the matcher with a depth of ten, which sets both the number of documents retrieved per source and the number of people returned, and match gold researchers by case-folded name, because the retrieval output carries no repository identifier. We report  $\text{Hit}@k$ , the fraction of queries with a gold researcher in the top  $k$ , and the mean reciprocal rank (MRR) [15] of the first gold researcher within the top ten, with a miss counted as zero. **[record the matcher revision, parser and retrieval settings used for the evaluation]**

The benchmark has known limits. Each query is derived from its target’s own documents, so it tests whether THEMIS recovers the authors of a text, and the queries probably share more vocabulary with their sources than a student’s request would. Other researchers who would supervise well count as misses. Generated test collections can support retrieval evaluation [13], but we do not know how closely these requests resemble real student queries. We therefore read the results as a measure of retrieval and crediting, not of recommendation quality, which needs judgements from students and supervisors.

### 5.2 Results

Table 2 gives the results. A gold researcher is ranked first for 66.0% of queries and appears in the top five for 83.2%. Posting queries score higher than publication queries (MRR 0.841 against 0.619). A posting names its supervisor directly and sits among 695 documents, whereas a publication query must single out its author among 214,756 records, each of which may credit several UZH co-authors. These are subsets of one benchmark, not a comparison of single-source systems. MRR rises with the number of keywords requested, as expected when queries echo their source text. Publication targets

**Table 2: Ranking results on 900 generated queries. Rows below the first break results down by source, by the number of keywords requested, and, for publication queries, by the target’s number of publications with abstracts.**

| Subset              | <i>n</i> | Hit@1 | Hit@5 | Hit@10 | MRR  |
|---------------------|----------|-------|-------|--------|------|
| All queries         | 900      | .660  | .832  | .848   | .730 |
| Publication queries | 450      | .529  | .751  | .762   | .619 |
| Posting queries     | 450      | .791  | .913  | .933   | .841 |
| Up to 2 keywords    | 300      | .593  | .817  | .837   | .684 |
| Up to 4 keywords    | 300      | .663  | .810  | .833   | .724 |
| Up to 8 keywords    | 300      | .723  | .870  | .873   | .782 |
| ≥ 10 publications   | 267      | .491  | .738  | .749   | .590 |
| < 10 publications   | 183      | .585  | .770  | .781   | .661 |

**Table 3: Query variants generated with and without information about the student (100 publication targets).**

| Query content                  | Hit@1 | Hit@5 | MRR  |
|--------------------------------|-------|-------|------|
| Topic only                     | .450  | .730  | .563 |
| Topic + study programme        | .470  | .670  | .559 |
| Topic + background             | .410  | .680  | .527 |
| Topic + programme + background | .410  | .670  | .521 |

with ten or more publications with abstracts are found less often than those with fewer (MRR 0.590 against 0.661), a difference we have not yet explained.

### 5.3 Topic Specificity

We ran an exploratory probe of whether rankings favour a researcher’s own topic over broader or neighbouring ones. For eight researchers we compared three queries, one generated from their publications, one naming a neighbouring topic that we chose, and one naming the general field. The own-topic query ranked the researcher above both others for four of the eight. Two researchers did not reach the top ten for any of the three queries. For the remaining two, a neighbouring or general query ranked the researcher higher than their own-topic query, and in one of these the generated own-topic query itself mentions the neighbouring topic, which confounds the comparison. With eight cases and neighbouring topics nobody has judged, we treat these results as a prompt for a proper test rather than as a measurement.

### 5.4 Student Background

Students often describe more than a topic, adding their degree programme, courses or earlier projects. For 100 publication targets we generated four query variants from the same keyword set: topic only, with the study programme, with the background, and with both (Table 3). The variants with student context had lower MRR and Hit@5, although adding the programme raised Hit@1 from 0.450 to 0.470. Each variant was generated separately, so its wording changed along with the context, and the comparison does not isolate the effect of the context itself. A controlled test would add fixed context to a single base query and report paired uncertainty.

## 6 Conclusion

THEMIS recommends supervisors from a university’s own publication records and thesis postings, links each recommendation to its evidence, and resolves people across the two sources with a deliberately strict rule. On generated queries a gold researcher appears in the top five for 83.2% of cases. The probes point to weaknesses in separating a specialisation from its field, and the evidence on adding student context is inconclusive.

Several parts remain open. We have not compared THEMIS with single-source retrieval or with a lexical expert-finding baseline, so the benefit of combining the sources is untested. Ranking uses only the best raw similarity, although publication counts, recency and postings are available, and the per-source thresholds rest on nine probe queries. Identity resolution reaches a quarter of supervisors, and the researcher profiles already scraped from department pages are a likely source for many of the rest. The integration into askUZH is not yet deployed. The central open question is whether the recommendations help students, which is for the study with students and supervisors that we are preparing.

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[thank the supervisors and the askUZH team by name]

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